

Evaluation of Autoinjector Safety Events in Health-System Specialty Pharmacy Patients

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BACKGROUND

Autoinjector medications are a cornerstone of medical management in multiple chronic disease states, including inflammatory bowel disease, hypercholesterolemia, migraines, and rheumatoid arthritis. In these disease states, successful medication administration is crucial to prevent exacerbations or disease progression. Medication delivery via self-injection provides an effective and convenient option for patients¹. In 2024, three of the top five drugs by expenditure in the United States were self-administered injections, including semaglutide, adalimumab, and dulaglutide, while all the top five specialty medications by expenditure were all self-injectable (adalimumab, ustekinumab, dupilumab, risankizumab, and etanercept)². Despite prior studies showing that specialty pharmacy interventions improve patient outcomes and adherence, at the University of Kentucky Specialty Pharmacy and Infusion Services (UKSPIS) there are still opportunities to reduce the risk of auto-injector misfires. This study aimed to identify trends in self-injectable misfire events to develop a targeted educational intervention to reduce avoidable misfires.

METHODS

- Study Design**
- Retrospective, observational study
 - Auto-injector dispensing history and patient demographics (age, ethnicity, gender, payor, preferred language, and race) were collected by chart review of Epic and Therigy
- Inclusion Criteria**
- Patients who received a prescription dispense for a self-injectable medication between 01/01/2022-12/31/2022 at UKSP
- Exclusion Criteria**
- Patients ≤ 18 years old were excluded due to confounding influence of administration by parents or guardians
- Primary Outcome**
- Incidence of misfire events per 1,000 fills
- Secondary Outcomes**
- Odds of misfire in comparison to other auto-injector medications
 - Demographic factors associated with misfires
 - Whether the misfire resulted in a delay in care
- Statistical Analysis**
- Incidence rate per fill
 - with the numerator being the number of misfires and denominator being number of fills
 - Incidence rate per 100 patients
 - with the numerator being the number of misfires and denominator being the number of patients
 - Odds ratio for medication-specific misfire risk in comparison to other autoinjector dispenses using univariable logistic regression

RESEARCH QUESTION

What factors influence the occurrence of safety events in patients who fill autoinjector medications at the University of Kentucky Specialty Pharmacy?

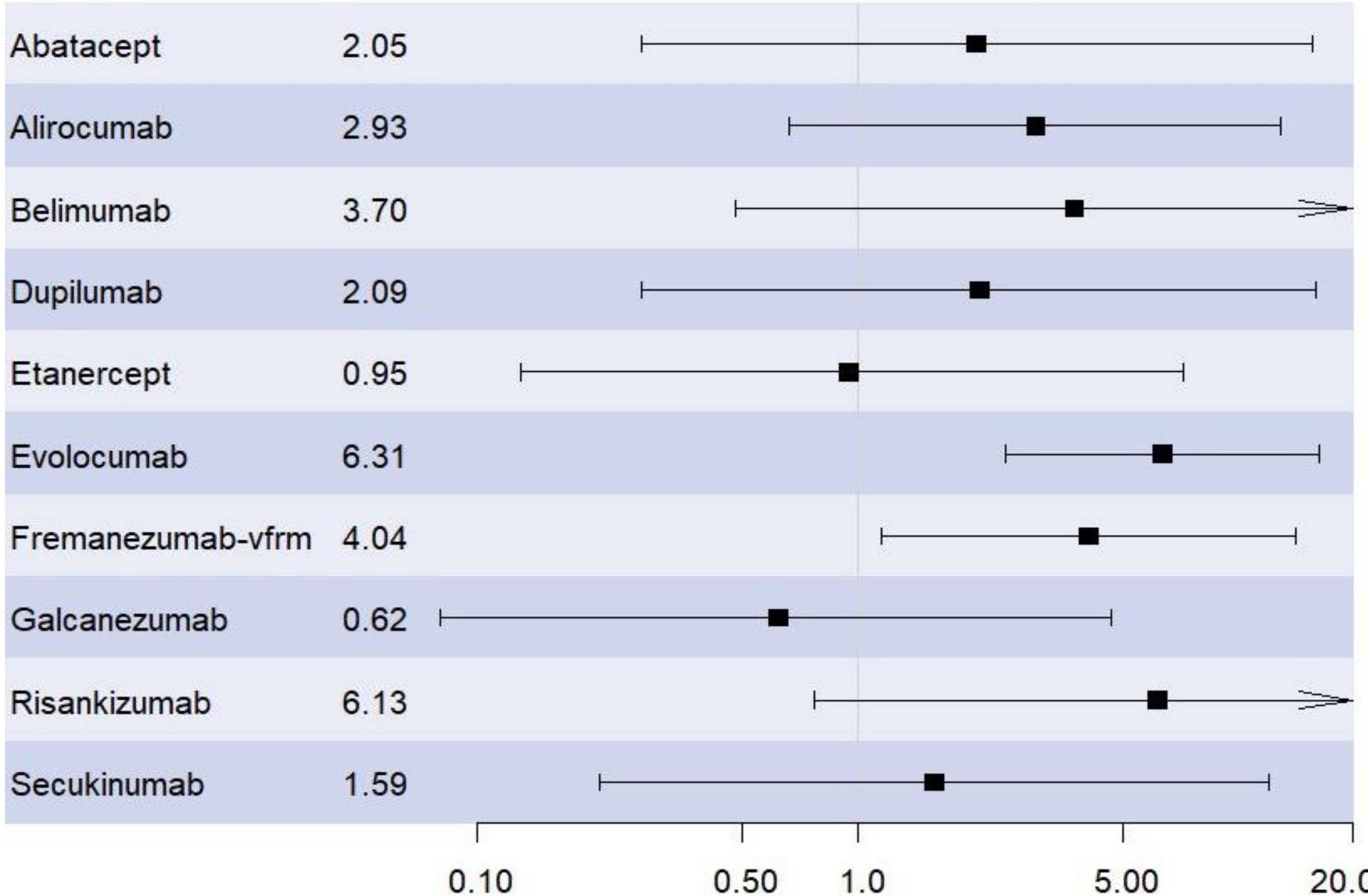
Table 1. Patient Demographics

Attribute		No Reported Misfires (n=1,892)	Misfire reported (n=19)	P-value
Age	Mean (SD)	50.5 (15)	53.8 (18.7)	0.2343
Sex	Female	1,367 (72.3%)	13 (68.4%)	0.614
Ethnicity	Hispanic, Latino/a, or Spanish Origin	30 (1.6%)	0	1
	Not Hispanic, Latino/a, or Spanish Origin	1,788 (94.5%)	19 (100%)	
Preferred Language	English	1,861 (98.4%)	19 (100%)	1
Payor	Cash/ Self-Pay	9 (0.5%)	0	0.1023
	Commercial/ Private	675 (35.7%)	2 (10.5%)	
	Medicaid	604 (31.9%)	8 (42.1%)	
	Medicare Part D	602 (31.8%)	9 (47.4%)	

Table 2. Incident Rates

Generic	Number of fills	Number of Incidents	Incidents per 1000 fills	Incidents per 100 Patients	Odds Ratio (95% CI)	P-value
Abatacept	317	1	3.15	1.96	2.05 (0.27,15.63)	0.49
Alirocumab	407	2	4.91	2.67	2.93 (0.66,12.93)	0.16
Belimumab	188	1	5.32	3.45	3.70 (0.48, 28.67)	0.21
Dupilumab	380	1	2.63	2.00	2.09 (0.27,15.97)	0.48
Etanercept	740	1	1.35	0.94	0.95 (0.13,7.15)	0.96
Evolocumab	1067	7	6.56	4.19	6.31 (2.45, 16.26)	0.00
Fremanezumab-vfrm	532	3	5.4	3.45	4.04 (1.15,14.12)	0.03
Galcanezumab	1020	1	0.98	0.64	0.62 (0.08,4.66)	0.64
Risankizumab	43	1	23.26	5.56	6.13 (0.77,48.54)	0.09
Secukinumab	462	1	2.16	1.54	1.59 (0.21, 12.07)	0.66

Figure 1. Forest Plot- Odds of Misfire Events



RESULTS

- Evolocumab [6.56 incidents per 1,000 fills; OR 6.31, 95%CI 2.45-16.26] and fremanezumab [5.64 incidents per 1,000 fills; OR 4.04, 95%CI 1.15-14.12] were identified as having significantly higher risk of misfire as compared to other autoinjector medications
- 42.1% of misfires resulted in a delay in therapy
- No demographic factors were significantly associated with misfire events (n=19)

CONCLUSION

- Lack of significance in demographic association suggests that misfires may be a result of inadequate patient education or potential issues with device design
- Need for proactive patient education on autoinjector medications, especially neurology and primary care patients
- Benefit of applying the specialty care model to our neurology and primary care patients receiving nonspecialty medications with complex administration requirements, increasing patient counseling and patient follow ups

CITATIONS

- Eric M Tichy, James M Hoffman, Mina Tadrus, Matthew H Rim, Sandra Cuellar, John S Clark, Mary Kate Newell, Glen T Schumock. National trends in prescription drug expenditures and projections for 2024. *American Journal of Health-System Pharmacy*. Volume 81, Issue 14, 15 July 2024, Pages 583–598, <https://doi.org/10.1093/ajhp/zzae109>
- Xiao X, Li W, Clawson C, Karvani D, Sondag P, Hahn JK. Evaluation of performance, acceptance, and compliance of an auto-injector in healthy and rheumatoid arthritic subjects measured by a motion capture system. *Patient Prefer Adherence*. 2018;12:515-526. doi:10.2147/PPA.S16039